

Do Recurrent RL Agents Discover a Unique Internal Clock?

Evidence Against a Canonical Temporal Code

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Keywords: temporal representation, recurrent neural networks, reinforcement learning, representational analysis, CKA

Summary

We examine how LSTM-based RL agents frame temporal information in their hidden states when no explicit clock is provided. Using CKA comparison with engineered encodings, linear probing, and PCA, we find that temporal encoding is training-dependent, non-unique across random seeds, and sensitive to temporal horizon. These findings challenge the implicit assumption that recurrent networks discover a single canonical temporal code, and suggest that the framing of time in learned representations is contingent on optimization path and task structure.

Contribution(s)

1. We show that temporal encoding in recurrent RL agents is not unique—different training runs produce different internal clocks despite identical behavior.
Context: Challenges the implicit “canonical temporal code” assumption in RL representation analysis.
2. We demonstrate that the representational geometry shifts with temporal horizon, from sinusoidal-like to scalar-like encoding.
Context: Links task structure to the learned framing of time, with implications for transfer and generalization.
3. We provide a simple, controlled methodology (CKA against engineered codes + multi-seed comparison) for testing uniqueness assumptions about learned representations.
Context: Applicable beyond time to other latent task variables.

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Abstract

1 Recurrent RL agents without an explicit clock must invent their own internal representa-
 2 tion of time. A natural assumption is that this representation is unique: any sufficiently
 3 trained agent converges to the same canonical temporal code. We show this assump-
 4 tion does not hold. LSTM agents trained on the same periodic task construct different
 5 temporal codes depending on their random initialization, despite identical behavioral
 6 performance. Using Centered Kernel Alignment (CKA) against engineered temporal
 7 encodings, linear probing, and PCA on a periodic gridworld, we ask three questions:
 8 (Q1) Is temporal structure learned or inherent to recurrence? (Q2) Is the emergent en-
 9 coding unique across training runs? (Q3) How does the encoding depend on the tempo-
 10 ral horizon? Untrained LSTM controls show weak temporal structure, which emerges
 11 reliably after RL training. Multi-seed experiments reveal that the learned encoding
 12 is *non-unique*: the highest-CKA engineered encoding varies across seeds, from sinu-
 13 soidal positional encoding to phase-based encoding. A controlled horizon manipulation
 14 shows that the encoding is also *horizon-sensitive*: at short horizons, agents align with
 15 sinusoidal codes; at longer horizons, alignment shifts toward scalar encoding. These
 16 findings hold implications for how we interpret recurrent representations: if there is no
 17 single canonical temporal code, single-seed representational analyses may not general-
 18 ize.

19 1 Introduction

20 RL researchers typically assume the hidden state of a recurrent network is a unified representation of
 21 the agent’s situation—including elapsed time. But when no clock signal is provided, the agent has to
 22 *construct its own framing* of the temporal dimension. This invites a question: given a well-defined
 23 POMDP, do trained agents converge to a *canonical* temporal code—a unique, reproducible internal
 24 clock? The answer is no: different training runs produce different temporal codes despite identical
 25 behavioral performance.

26 Prior work has established that time cells emerge in RNN-based RL agents, analogous to hippocam-
 27 pal time cells (MacDonald et al., 2011; Lin et al., 2023). In NLP, the form of positional encoding—
 28 sinusoidal, learned, or relative—has been a central design choice in the Transformer architecture
 29 (Vaswani et al., 2017). The *variability and contingency* of such representations in recurrent RL
 30 agents has received less attention.

31 We study three questions:

32(Q1) Does temporal structure emerge through training, or is it inherent to recurrent architectures?

33(Q2) Is the emergent encoding unique across training runs?

34(Q3) How does the encoding depend on the temporal horizon of the task?

We study these questions by training LSTM agents on DoorGrid, an environment with periodic dynamics, and comparing their hidden states against a library of engineered temporal encodings using Centered Kernel Alignment (CKA) (Kornblith et al., 2019), complemented by linear probing, PCA, and single-neuron phase selectivity analysis.

The temporal framing learned by LSTM agents is *non-unique*: agents with identical task performance align differently with simple engineered temporal codes, and the preferred alignment shifts with temporal horizon. This challenges how we interpret and compare recurrent representations across training runs—if there is no single canonical code, what exactly are we comparing?

2 Related Work

2.1 Temporal Representations in RL

Lin et al. (2023) showed that time cells emerge in RNN-based RL agents, extending the neuroscience observation that hippocampal neurons encode elapsed time (MacDonald et al., 2011). Hausknecht and Stone (2015) introduced DRQN, showing that recurrent connections enable RL agents to handle partial observability. Kapturowski et al. (2019) scaled this with R2D2 using LSTM-based agents on Atari. Jaderberg et al. (2017) showed that auxiliary temporal prediction tasks improve representation learning. Our work examines not just *whether* agents encode time, but whether this encoding has a unique canonical form.

2.2 Probing Neural Representations

Linear probing (Alain and Bengio, 2017) and CKA (Kornblith et al., 2019) are standard tools for analyzing representations. Hewitt and Manning (2019) emphasized that probe accuracy alone is insufficient—one must control for selectivity. Voita and Titov (2020) studied information-theoretic aspects of probing. CKA is used here as a descriptive similarity measure to a small library of candidate temporal codes, not as an identification procedure for the true latent code.

2.3 Multiple Solutions in Neural Networks

The existence of multiple loss landscape basins is well documented (Garipov et al., 2018; Li et al., 2018; Draxler et al., 2018). In neuroscience, Edelman and Gally (2001) formalized *degeneracy*—distinct structures performing the same function. We ask a question that sits at the intersection of these lines of work: do different optimization basins produce qualitatively different *temporal representations* for the same RL task?

3 Methods

3.1 Environment

DoorGrid is an 8×8 grid with a periodic door that alternates between open and closed. At $T = 4$, the door is open during phases 0–1 and closed during phases 2–3. The agent receives a one-hot encoding of its (x, y) position (64-dimensional); no explicit temporal features are provided. Actions are {up, down, left, right}. The reward is +1 for reaching the goal (top-right corner), −1 for stepping into a hazard (a cell that is terminal when the door is closed), and 0 otherwise. Episodes terminate upon reaching the goal, hitting a hazard, or exceeding a step limit. For horizon scaling, we vary $T \in \{4, 8, 16, 32\}$; the door is open during the first half of each period and closed during the second half.

This minimal environment isolates temporal encoding from complex spatial navigation: the periodic structure is the agent’s primary challenge.

Table 1: Trained vs. untrained LSTM on DoorGrid ($T = 4$). $R^2(t)$: timestep prediction. $\text{Acc}(\phi)$: phase classification. CKA Best: highest CKA with any engineered encoding. Return: best evaluation reward (20-episode mean). Trained row shows seed 42 as representative; all 5 seeds achieve $R^2 = 1.0$ and Return = 0.920. Untrained LSTM does not participate in the RL task, so Return is not applicable.

Condition	$R^2(t)$	$\text{Acc}(\phi)$	CKA Best	MI norm	Return
Trained (seed 42)	1.000	1.000	0.740	1.000	0.920
Untrained	0.304	0.298	0.153	0.082	N/A

3.2 Agents and Training

A single-layer LSTM ($d = 64$) with linear policy and value heads, trained with REINFORCE with baseline (learning rate 10^{-3} , entropy coefficient 0.01, discount $\gamma = 0.99$). An MLP baseline (2-layer, 64 units each) processes each observation independently, serving as a sanity check for the necessity of recurrence. All experiments use 5 random seeds $\{42, 123, 456, 789, 1024\}$.

3.3 Analysis Pipeline

After training, we run 200 evaluation episodes and collect hidden state vectors $\mathbf{h}_t \in \mathbb{R}^{64}$ along with timestep t and phase $\phi = t \bmod T$.

Linear Probe. Ridge regression from $\mathbf{h}_t$ to t (R^2 score) and logistic regression to ϕ (classification accuracy), with shuffled-target controls ($R_{\text{shuffled}}^2 = 0.020$, confirming no leakage).

PCA. PC1 variance fraction and its correlation with timestep.

Mutual Information. Normalized mutual information $I(h_d; \phi)/H(\phi)$ between each hidden dimension and phase, averaged over dimensions, quantifying how much phase information is distributed across the representation.

CKA. Linear CKA (Kornblith et al., 2019) between $\mathbf{h}_t$ and engineered temporal encodings: raw scalar (t/T), sinusoidal PE ($\sin / \cos$ of $2\pi t/T$), phase $\sin/\cos$, one-hot phase, and remaining time $((T - t)/T)$. As reference baselines, CKA between engineered encodings themselves ranges from 0.38 to 0.54 at $T = 8$, and CKA between any encoding and shuffled hidden states is approximately 0.

Single-Neuron Phase Selectivity. For each hidden dimension d , we compute the mean activation per phase $\bar{h}_d(\phi)$ and a selectivity index $s_d = (\max_{\phi} \bar{h}_d - \min_{\phi} \bar{h}_d) / (\max_{\phi} \bar{h}_d + |\min_{\phi} \bar{h}_d| + \epsilon)$. Neurons with $s_d \approx 1$ are strongly phase-tuned. We compare the top- k phase-selective dimensions across seeds.

4 Results

4.1 Temporal Structure Emerges Through Training (Q1)

Table 1 speaks to Q1. Trained LSTMs show strong temporal encoding ($R^2 = 1.0$, CKA = 0.74, phase accuracy 1.0) while untrained LSTMs show CKA ≈ 0 and phase accuracy at chance (0.30 for 4 classes). Temporal structure is training-dependent, not an artifact of random recurrent dynamics. The MLP baseline (which lacks recurrence) fails to learn the task (Return = -2.0 across all 5 seeds, $R^2(t) \approx 0$), confirming that temporal tracking requires recurrent state.

Table 2: CKA (mean $\pm$ std across 5 seeds) between DoorGrid ($T = 4$) hidden states and engineered encodings. High std relative to mean indicates seed-dependent encoding. Highest-CKA mean in **bold**.

Encoding	CKA (mean $\pm$ std)
Raw scalar (t/T)	0.40 ± 0.10
Sinusoidal PE	0.63 ± 0.09
Phase sin/cos	0.28 ± 0.21
One-hot phase	0.27 ± 0.16

Table 3: Representation similarity shifts with temporal horizon in DoorGrid. CKA (mean $\pm$ std, 5 seeds) with engineered temporal encodings. [†]1 seed failed at $T = 16$ and 2 at $T = 32$; stats computed on all 5. Converged-only: $T = 16$: CKA(raw)=0.59, sin_PE=0.65; $T = 32$: CKA(raw)=0.61, sin_PE=0.54.

T	CKA(raw)	CKA(sin_PE)	PC1-t	$R^2(t)$	Return
4	0.40 ± 0.10	0.63 ± 0.09	0.44 ± 0.10	1.000	0.920
8	0.56 ± 0.14	0.58 ± 0.08	0.46 ± 0.29	1.000	0.880
16	0.64 ± 0.15	0.60 ± 0.10	0.63 ± 0.37	1.000	0.800 [†]
32	0.70 ± 0.22	0.50 ± 0.08	0.71 ± 0.28	1.000	0.640 [†]

106 4.2 The Temporal Framing Is Not Unique (Q2)

107 All 5 seeds achieve $R^2 = 1.0$, phase accuracy = 1.0, and Return = 0.920—functionally identical
 108 performance. Yet the CKA profiles differ substantially across seeds. Of 5 seeds, 4 show high-
 109 est CKA with sinusoidal PE (0.64–0.74), while seed 456 shows highest CKA with phase sin/cos
 110 (0.62) with sinusoidal PE scoring lower (0.46). The coefficient of variation across seeds is 0.15
 111 for CKA(sin_PE) and 0.76 for CKA(phase), indicating that the encoding geometry is sensitive to
 112 random initialization. Cross-seed CKA between the agents’ own hidden states (aligned by timestep)
 113 is uniformly high (> 0.96), so the overall representational geometry is similar across seeds. The
 114 plurality we identify is not in the raw hidden-state structure but in the *alignment with specific en-*
 115 *gineered codes*: the same population geometry can be more or less similar to different candidate
 116 temporal codes depending on initialization.

117 **Single-neuron evidence.** We identify the 8 hidden dimensions with highest phase selectivity ($s_d \approx$
 118 1.0 for all seeds) as “time cells.” Across all 10 pairs of seeds, the mean overlap in top-8 time cell
 119 dimensions is 1.8/8 (mean Jaccard index = 0.14). However, with 34 of 64 dimensions showing
 120 near-maximal selectivity ($s_d > 0.9$), the hypergeometric expectation for random overlap is $8 \times$
 121 $8/34 \approx 1.88$ —making the observed overlap indistinguishable from chance. We therefore do not
 122 treat single-neuron overlap as evidence for or against representational plurality; the population-level
 123 CKA differences (above) provide the stronger signal. Because hidden-unit indices are not aligned
 124 across independently trained models, we treat the overlap analysis as descriptive (see Appendix A
 125 for per-seed details).

126 4.3 The Framing Shifts with Temporal Horizon (Q3)

127 Table 3 and Figure 1 answer Q3. At $T = 4$, mean similarity to sinusoidal PE is highest (CKA
 128 = 0.63 vs. raw scalar = 0.40). At $T = 8$, the two are roughly equal (CKA ≈ 0.57). At $T \geq 16$,
 129 similarity to raw scalar exceeds sinusoidal PE. The |PC1-timestep correlation| also increases with
 130 period length (from 0.44 at $T = 4$ to 0.71 at $T = 32$), indicating that temporal information becomes
 131 more concentrated in a single dominant dimension at longer horizons.

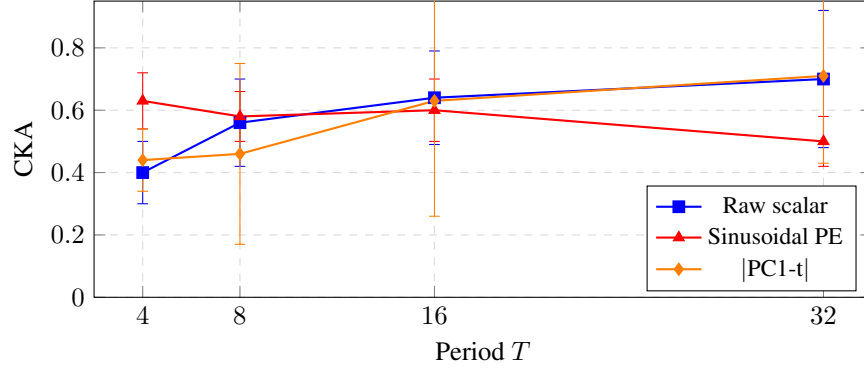


Figure 1: CKA with engineered encodings and lPC1-timestep correlation as a function of period T (mean $\pm$ std across 5 seeds). Similarity to raw scalar encoding increases at longer horizons, while similarity to sinusoidal PE decreases.

132 The seed-level variability persists across all period lengths. At $T = 8$, seed 42 shows highest CKA
133 with raw scalar (0.81) while seeds 123/456/789 show highest CKA with sinusoidal PE (0.48–0.58).
134 At $T = 16$, seed 1024 matches raw scalar (0.84) while seed 789 matches remaining time (0.77).
135 The framing of time is not an artifact of a single period length.

136 At $T = 32$, training becomes less stable: 2 of 5 seeds fail to converge, and task return drops to 0.640
137 (see Appendix B for failed-seed training curves). The CKA shift at long horizons may partly reflect
138 optimization difficulty rather than purely representational adaptation.

139 5 Discussion

140 LSTM-based RL agents do not converge to a single canonical temporal code. The code they learn
141 depends on their initialization, and which engineered encoding it resembles shifts with the task’s
142 temporal horizon. These findings hold implications for how we interpret recurrent representations.

143 **No canonical internal clock.** The assumption that recurrent networks provide a unified, well-
144 defined temporal code is embedded in much of RL practice—for example, when comparing repre-
145 sentations across seeds or architectures, or when interpreting hidden states as encoding specific task
146 variables. Our results indicate that, at least in simple periodic environments, this assumption does
147 not hold: the framing of time is contingent on optimization path and task structure. This has direct
148 methodological implications: a study using only seed 42 would conclude “LSTMs encode time via
149 sinusoidal-like codes,” while a study using only seed 456 would conclude “LSTMs encode time via
150 phase-based codes.” Neither generalizes.

151 **Interpretability.** The representational plurality we observe raises a question for RL interpretability:
152 if the temporal code is not unique, what does it mean to “interpret” a neuron as a time cell? The same
153 functional role (phase discrimination) is implemented by different neuronal subpopulations across
154 seeds. This matches Edelman and Gally (2001)’s concept of degeneracy—structurally different
155 elements performing the same function—and suggests that single-seed interpretability analyses may
156 be misleading.

157 **What does “representation” mean in RL?** If temporal codes are non-unique, then “what the agent
158 has learned” is not a well-defined mapping from task to internal code, but one of many functionally
159 equivalent solutions. This is Edelman’s *degeneracy* (many-to-one structure-to-function mapping),
160 not mere *redundancy* (duplicate copies of the same structure): the solutions differ qualitatively, not
161 just numerically. If the code is path-dependent, then any single-seed analysis characterizes one
162 solution among many, not a canonical internal structure. Whether this plurality is a nuisance for
163 interpretability or a genuine feature of learning systems is open.

164 **Scope and limitations.** We do not claim a universal taxonomy of temporal coding in RL. Instead, we
 165 isolate one setting where temporal structure can be controlled and representation geometry inspected
 166 directly. Three caveats: First, our experiments use one environment, one algorithm (REINFORCE),
 167 and one architecture (single-layer LSTM). Second, at $T = 32$, training instability (2/5 seeds fail)
 168 limits the horizon analysis. Third, CKA measures linear similarity and may miss nonlinear encoding
 169 strategies. Whether representational plurality persists across architectures (GRUs, attention-based
 170 models) and in more complex domains remains to be tested.

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217 A Single-Neuron Phase Selectivity Analysis

Table 4: Top-8 time cell dimensions and preferred phases per seed (DoorGrid, $T = 4$). Mean pairwise overlap: 1.8/8 dimensions (Jaccard = 0.14). Hypergeometric expectation given 34/64 selective dims: ≈ 1.88 .

Seed	Top-8 dims	Preferred phases
42	28, 43, 7, 44, 31, 32, 50, 39	2, 2, 0, 1, 3, 1, 3, 2
123	2, 40, 25, 49, 14, 4, 33, 36	1, 2, 3, 3, 3, 1, 3, 2
456	40, 35, 44, 39, 36, 22, 26, 43	3, 2, 0, 0, 3, 2, 3, 2
789	25, 51, 10, 15, 29, 18, 24, 38	1, 1, 1, 3, 3, 3, 3, 3
1024	43, 10, 25, 11, 39, 4, 36, 40	3, 1, 2, 3, 2, 1, 3, 2

- 218 Hidden-unit indices are not aligned across independently trained models, so low overlap in unit
219 indices is partially expected under permutation invariance. With 34 of 64 dimensions showing
220 high phase selectivity, the observed overlap (1.8/8) is close to the hypergeometric chance level
221 (≈ 1.88), so this analysis does not provide evidence for or against representational uniqueness. The
222 population-level CKA differences provide the stronger signal: seed 42 has $\text{CKA}(\text{sin_PE}) = 0.74$
223 while seed 456 has $\text{CKA}(\text{phase}) = 0.62$ with $\text{CKA}(\text{sin_PE}) = 0.46$, indicating genuinely different
224 encoding geometries.

225 B Failed Seeds at Long Horizons

- 226 At $T = 16$, seed 1024 fails to converge (best eval reward = -2.0). At $T = 32$, seeds 456 and
227 789 fail to converge (best eval reward = -2.0). The reported CKA statistics in Table 3 include all
228 5 seeds, as even non-converged agents contain representational structure worth characterizing. At
229 $T = 32$, excluding the 2 failed seeds: $\text{CKA}(\text{raw}) = 0.61 \pm 0.28$ (vs. 0.70 ± 0.22 with all 5), and
230 $\text{CKA}(\text{sin_PE}) = 0.54 \pm 0.10$ (vs. 0.50 ± 0.08). The same pattern holds—raw scalar still beats sin
231 PE.